

Unit 4 – Veterinary Immunology & Serology

All 19 syllabus topics on five working sheets. Immunology is the unit where the same six diagrams – the antibody, the complement cascade, antigen presentation, the four hypersensitivities, the serological reactions and the vaccine table – answer most of the paper. Those six are drawn here in full.

PAPER	SYLLABUS TOPICS	REVISION TIME	COURSE
Paper II · Units 4, 5 Theory 100 · Practical 60	19 of 19 covered	25 min full pass 10 min sheets 03–05	B.V.Sc & A.H. II Yr VCI MSVE · 3+2=5

Sheet	What it covers	Syllabus topics folded in
01	History, lymphoid organs & cells, innate vs acquired immunity	t01–t03
02	Antigens, haptens, adjuvants, immunoglobulins, antibody diversity	t04–t08
03	MHC, BCR & TCR, antigen processing, complement, cytokines	t11–t15
04	Serological reactions and monoclonal antibodies	t09–t10
05	Hypersensitivity, tolerance & autoimmunity, antimicrobial immunity, vaccines	t16–t19

- Organs, cells & general

■ Antigen & antibody

■ Complement & cytokines

■ Hypersensitivity & tolerance

■ Serology & vaccines
- Exam trap & mnemonic

Innate vs acquired immunity

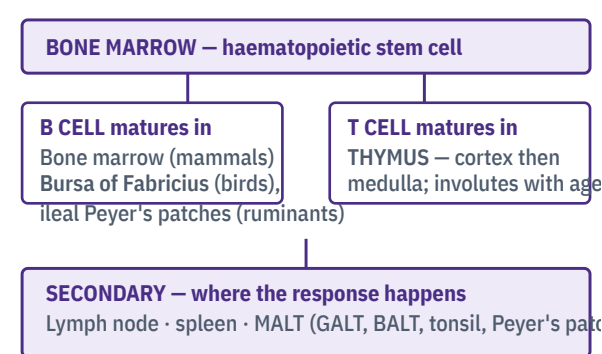
TABLE

Feature	Innate (natural, non-specific)	Acquired (adaptive, specific)
Present	From birth	Develops after exposure
Specificity	Broad — recognises PAMPs via PRRs (TLRs, NLRs)	Exquisitely specific for an epitope
Memory	None (trained immunity is a modern nuance)	Present — the basis of vaccination
Response time	Minutes to hours	Days (primary), hours to days (secondary)
Components	Skin & mucosa, cilia, lysozyme, low pH, normal flora, fever, complement (alternative & lectin) , interferon, defensins, NK cells, neutrophils, macrophages, inflammation	B cells & antibody (humoral), T cells (cell-mediated), APCs, cytokines
Improves on repeat	No	Yes

- **Classify acquired immunity** — always draw this 2 × 2:
Active natural = recovery from infection · **Active artificial** = vaccination · **Passive natural** = colostral/maternal antibody, transplacental · **Passive artificial** = antiserum, antitoxin, hyperimmune serum.
- **Active** — slow onset, **long-lasting**, **has memory**, needs a functioning immune system. **Passive** — immediate, **short-lived (2–3 weeks)**, **no memory**, useful in an emergency (tetanus antitoxin, snake antivenin).
- **Species point that gets asked:** the ruminant, horse and pig placenta is **epitheliochorial/syndesmochorial** — **no maternal IgG crosses in utero**, so the neonate is **agammaglobulinaemic at birth** and depends entirely on **colostrum** absorbed by pinocytosis in the first 6–12 h (gut "closure" by 24–36 h). In dogs and cats ~5–10% crosses the endotheliochorial placenta; in primates and rabbits (haemochorial) most crosses. **Failure of passive transfer** is the single commonest cause of neonatal septicaemia in calves and foals.
- **Other terms:** **herd immunity**, **local/mucosal immunity** (IgA), **species**, **breed** and **individual resistance**, **adoptive immunity** (transfer of sensitised lymphocytes).

Lymphoid organs

MECHANISM



Node: B cells in cortical follicles/germinal centres; T cells in the paracortex. **Spleen:** B in follicle, T in PALS

Two species facts examiners love: the **pig has an inverted lymph node** — medulla outside, cortex inside, with afferent lymph entering centrally; and the **bursa of Fabricius**, a dorsal cloacal sac, is the site of B-cell maturation unique to birds (bursectomy abolishes antibody production but leaves cell-mediated immunity intact — the experiment that separated the two arms). **Haemal nodes** occur in ruminants and filter blood rather than lymph.

Cells of the immune system

RAPID-FIRE

B lymphocyte	Surface Ig (BCR), MHC-II, CD19/CD20/CD21; → plasma cell (antibody factory, eccentric nucleus, cartwheel chromatin) and memory B cell
T helper (Th)	CD4 , recognises antigen on MHC-II; the master regulator
Th1 / Th2 / Th17 / Treg	Th1 → IFN- γ , IL-2 → cell-mediated immunity , macrophage activation · Th2 → IL-4, 5, 10, 13 → antibody & IgE , anti-parasite · Th17 → IL-17, neutrophil recruitment · Treg (CD4 ⁺ CD25 ⁺ FoxP3 ⁺) → IL-10, TGF- β , peripheral tolerance
T cytotoxic (Tc)	CD8 , recognises antigen on MHC-I; kills by perforin-granzyme and Fas-FasL ; the answer to "how are virus-infected cells killed?"
NK cell	Large granular lymphocyte, no antigen receptor ; kills cells that have lost MHC-I ("missing self") and mediates ADCC through CD16 (Fc γ RIII)
Macrophage	From monocyte; phagocytosis , antigen presentation , IL-1/IL-6/TNF- α ; alveolar, Kupffer, microglia, osteoclast
Dendritic cell	The most efficient professional APC and the only one that can

Milestones in immunology

RAPID-FIRE

Variolation	Ancient China & India — dried smallpox crusts
Edward Jenner, 1796	Cowpox → smallpox; the word vaccine from <i>vacca</i> , a cow. Father of immunology
Louis Pasteur	Attenuation; fowl cholera, anthrax and rabies vaccines
Elie Metchnikoff, 1883	Phagocytosis — cellular theory of immunity
Behring & Kitasato, 1890	Diphtheria & tetanus antitoxin — humoral theory; first Nobel in Medicine, 1901
Bordet	Complement and complement fixation
Landsteiner	ABO blood groups; haptens
Portier & Richet, 1902	Anaphylaxis
von Pirquet	The term allergy
Ehrlich	Side-chain theory ; "horror autotoxicus"
Burnet, 1957	Clonal selection theory — the accepted explanation
Medawar	Acquired immunological tolerance
Edelman & Porter	Structure of the antibody molecule
Köhler & Milstein, 1975	Hybridoma & monoclonal antibodies

Sheet 01 · 60-second self-test

RECALL

Prompt	Answer
Father of immunology	Edward Jenner
Discovered phagocytosis	Metchnikoff
Clonal selection theory	Burnet, 1957
Primary lymphoid organs	Thymus + bone marrow (bursa in birds)
B-cell maturation site in birds	Bursa of Fabricius
T-cell area of a lymph node	Paracortex
T-cell area of the spleen	PALS
Species with inverted lymph node	Pig
Most efficient professional APC	Dendritic cell
Kills cells lacking MHC-I	NK cell
Marker of Th cells / Tc cells	CD4 / CD8
Immunity from colostrum	Passive natural
Placenta type in ruminants	Syndesmochorial — no IgG transfer

	prime a naïve T cell; Langerhans cell in skin	Tonegawa, 1976	Somatic recombination — the genetic basis of antibody diversity
Neutrophil	First responder; respiratory burst, myeloperoxidase, NETs. In cattle it is the predominant blood leucocyte in the newborn but lymphocytes predominate in the adult ruminant	Jerne	Idiotypic network theory
Eosinophil	Parasites (major basic protein) and type I hypersensitivity	1980	WHO declares smallpox eradicated; rinderpest declared eradicated 2011
Mast cell & basophil	Bear FcεRI; degranulate → histamine, heparin, leukotrienes		

02 **Antigens, Adjuvants & Immunoglobulins**

t04–t08

Antigen vocabulary

RAPID-FIRE

Antigen	A substance that combines specifically with antibody or a lymphocyte receptor. Immunogen = an antigen that can also induce a response. All immunogens are antigens; not all antigens are immunogens
Epitope / determinant	The small part actually bound — 5–8 amino acids or sugars. Linear (sequential) vs conformational . Valency = number of epitopes per molecule
Paratope	The antigen-binding site of the antibody
Hapten	Low-molecular-weight, antigenic but NOT immunogenic alone ; becomes immunogenic when coupled to a carrier protein. Examples: penicillin, picryl chloride, drugs causing allergy. Landsteiner's classic work
Mitogen	Non-specifically activates whole populations of lymphocytes. T cells — PHA, Con A · B cells — LPS (endotoxin), pokeweed mitogen acts on both . Used in the lymphocyte transformation test to assess cell-mediated immunity
Superantigen	Bridges MHC-II to the TCR Vβ outside the peptide groove → activates up to 20% of all T cells → cytokine storm. TSST-1, staphylococcal enterotoxins, streptococcal pyrogenic exotoxin
Adjuvant	Enhances the response to an antigen without being antigenic itself — see next module

Factor	Rule for good immunogenicity
Foreignness	The more phylogenetically distant, the better — self is normally tolerated
Molecular size	>10 kDa good, >100 kDa excellent ; <5 kDa usually poor
Chemical complexity	Proteins > polysaccharides > lipids > nucleic acids; aromatic residues help
Degradability	Must be processable by APCs — D-amino acid polymers are not immunogenic
Dose & route	Optimal dose; too little or too much gives tolerance . Route order: intradermal > subcutaneous > intramuscular > intraperitoneal > intravenous > oral
Host factors	Species, breed, age, genotype (MHC haplotype — immune response genes), nutrition, stress

Special antigen classes

TABLE

Class	Features
T-dependent	Proteins . Need Th help. Give IgM then class switch to IgG , affinity maturation and memory . Most vaccine antigens
T-independent	Polysaccharides, LPS, polymeric flagellin . Repeating epitopes cross-link the BCR directly → IgM only, no switching, no affinity maturation, poor/no memory . Why plain polysaccharide vaccines are poor in the young — solved by conjugating them to a protein carrier
Heterophile (Forssman)	Identical or closely similar antigens in phylogenetically unrelated species . Basis of cross-reaction . Examples: Forssman antigen (guinea pig, horse, cat, dog, sheep RBC); Proteus OX strains ↔ Rickettsia (Weil-Felix) ; Salmonella ↔ human RBC ; E. coli O157 ↔ Brucella (false-positive brucellosis tests)
Blood group	Alloantigens on RBCs. Cattle have >11 systems (J system — soluble, acquired from plasma); sheep 7; horses 7 with the Aa and Qa factors causing neonatal isoerythrolysis in foals and mules ; dogs DEA 1.1 is the important one (a first transfusion is usually safe, a second may not be); cats have naturally occurring alloantibodies — type B queens mated to type A toms give neonatal isoerythrolysis in kittens, so cats MUST be blood-typed before the first transfusion ; pigs A-O system
Autoantigen	Self components normally sequestered — lens, sperm, thyroglobulin, myelin

The immunoglobulin molecule

MECHANISM

HEAVY chain

$\mu \gamma \alpha \delta \epsilon$ — defines the class V_L

LIGHT chain

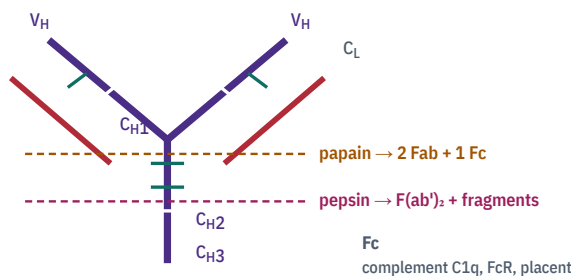
κ or λ only

Disulphide bonds

inter- and intra-chain

HINGE (proline-rich)

flexible; absent in IgM & IgE



- **Basic unit** = 2 identical heavy + 2 identical light chains = H_2L_2 , ~150 kDa, Y-shaped, held by disulphide bonds.
- **Fab** = fragment antigen binding, **monovalent** — cannot precipitate or agglutinate. **F(ab')₂** is **bivalent** and still cross-links. **Fc** = crystallisable, carries all the biological (effector) functions and the carbohydrate.
- **Variability terms:** **Isotype** = class/subclass differences (constant region, all members of a species) · **Allotype** = allelic differences between individuals · **Idiotypic** = differences in the variable region/paratope of one clone. **CDRs (hypervariable regions)**, three per chain, form the paratope.

The five classes

TABLE

Class	Chain · form	The point to remember
IgG	γ · monomer ~150 kDa	Most abundant in serum (~75%) and in most domestic-animal colostrum . Best at opsonisation , ADCC ; fixes complement (classical). The predominant antibody of the secondary response . Subclasses IgG1, IgG2 (IgG1 dominates cattle colostrum and milk)
IgM	μ · pentamer + J chain ~900 kDa	First antibody produced in a primary response and first synthesised by the foetus — so IgM in a precolostral serum sample proves intrauterine infection . 10 binding sites → best agglutinator and most efficient complement fixer ; too large to leave the vessels ("macroglobulin"). Also the monomeric BCR on naïve B cells
IgA	α · dimer + J chain + secretory component	The mucosal antibody — milk, saliva, tears, colostrum, respiratory, gut and urogenital secretions. Secretory component (from the epithelial poly-Ig receptor) protects it from proteolysis. Works by immune exclusion , does not fix complement classically. The predominant immunoglobulin in the milk of most species after the colostrum phase
IgE	ϵ · monomer lowest concentration	Binds FcεRI on mast cells and basophils → type I hypersensitivity and anti-parasite (helminth) immunity ; heat-labile (56 °C, 30 min) ; mediates PCA (Prausnitz-Küstner) reaction
IgD	δ · monomer	Chiefly a BCR co-receptor on mature naïve B cells ; trace amounts in serum; not found in cattle, pigs or chickens

Birds: only **IgM, IgY (the IgG equivalent) and IgA**; IgY is transferred to the **egg yolk** and IgA/IgM to the albumen — this is the basis of **maternal antibody in chicks** and of the **IgY technology** for producing antibodies from eggs.

Adjuvants

TABLE

Type	Examples & use
Mineral salts	Aluminium hydroxide / phosphate (alum), potassium alum — the classic; used in HS and clostridial bacterins; promotes Th2/antibody
Oil emulsions	Freund's incomplete (water-in-oil); Freund's complete = FIA + killed <i>M. tuberculosis</i> — the most powerful but too irritant for field use, laboratory only. Montanide, oil-adjuvant FMD and ND vaccines
Bacterial products	MDP, LPS/MPL, CpG oligodeoxynucleotides, cholera toxin B subunit, BCG
Surface-active	Saponin, Quil A , ISCOMs (immune-stimulating complexes)
Cytokines	IL-2, IL-12, IFN- γ , GM-CSF as molecular adjuvants
Particulate carriers	Liposomes, microspheres, nanoparticles, virus-like particles

Antibody production & diversity

RAPID-FIRE

Side-chain theory	Ehrlich — cells bear pre-formed receptors; antigen selects and the cell overproduces them. Partly correct in spirit
Template (instructive) theories	Breml & Haurowitz, Pauling — antigen moulds the antibody. Disproved: denatured antibody refolds with the same specificity, and antibody is made after antigen is cleared
Natural selection theory	Jerne, 1955 — antigen selects a pre-existing natural antibody
Clonal selection theory	Burnet, 1957 — the accepted one. Each lymphocyte is pre-committed to one specificity; antigen selects the matching clone, which proliferates into plasma cells and memory cells; self-reactive clones are deleted in fetal life ("forbidden clones")

Primary vs secondary response

TABLE

Feature	Primary	Secondary (anamnesic)
Lag phase	Long, 5–10 days	Short, 1–3 days
Peak titre	Low	10–100× higher
Main class	IgM	IgG (IgA at mucosa)
Affinity	Low	High — affinity maturation
Duration	Short	Long
Cell responding	Naïve B cell	Memory B cell
Antigen dose needed	High, adjuvant helps	Low

Mechanism – four points 1 Depot effect, slow antigen release · 2 Recruits and activates APCs at the site · 3 Increases effective antigen size and particulate presentation · 4 Directs the **type** of response (Th1 vs Th2). **Disadvantages:** injection-site granuloma and abscess, carcass condemnation, pain, and accidental self-injection of oil adjuvant is a surgical emergency in man.

Generation of diversity

Tonegawa's somatic recombination – V(D)J rearrangement by RAG-1/RAG-2 (heavy chain uses V-D-J, light chain V-J) · **combinatorial** pairing of any H with any L · **junctional** diversity and N-nucleotide addition by TdT · **somatic hypermutation** in the germinal centre giving **affinity maturation**. Together these produce $>10^9$ specificities from a few hundred genes

Class switching

Same V region joins a different C_H gene → specificity unchanged, **effector function changed**; driven by cytokines and CD40-CD40L

Idiotypic network

Jerne — anti-idiotypic antibodies regulate the response; the internal image can act as a vaccine

This table is the entire rationale for the booster dose — say that sentence when you draw it. It also explains why **paired sera taken 2–3 weeks apart showing a four-fold rise in titre** is the standard serological proof of recent infection.

03 MHC, Receptors, Antigen Presentation, Complement & Cytokines

t11–t15

MHC & antigen presentation — one diagram, two pathways

MECHANISM

ENDOGENOUS → MHC CLASS I

Viral / tumour protein made *inside* the cell
↓ ubiquitin → **PROTEASOME** → peptide 8–10 aa
↓ TAP transporter into the ER
↓ loaded on MHC-I (α chain + β₂-microglobulin)
↓ surface → presented to **CD8 Tc** → kill

EXOGENOUS → MHC CLASS II

Bacterium taken up from *outside*
↓ endosome → **LYSOSOME** → peptide 13–18 aa
↓ MHC-II arrives with the invariant chain (CLIP)
↓ HLA-DM swaps CLIP for the peptide
↓ surface → presented to **CD4 Th** → help

MHC CLASS I

On ALL nucleated cells (+ platelets)
α1 α2 α3 + β₂-m · groove closed at both ends
Binds **CD8** · target of Tc and of graft rejection
Species names: BoLA cattle · SLA pig · ELA horse
DLA dog · OLA sheep · B complex chicken · HLA man

MHC CLASS II

Only on professional APCs — dendritic cell, macrophage, B cell (+ thymic epithelium)
α + β chain · groove open at both ends
Binds **CD4** · Class III = C4, C2, factor B, TNF — not a presenting molecule at all

MHC is polygenic and the most polymorphic gene system known, is co-dominantly expressed, and is inherited as a haplotype block. Consequences to quote: it determines graft acceptance or rejection, it sets the **immune response (Ir) gene** control of vaccine responsiveness, and MHC associations underlie several autoimmune and disease-resistance traits (for example BoLA associations with mastitis resistance).

BCR and TCR compared

TABLE

Feature	BCR	TCR
Structure	Membrane IgM/IgD (H ₂ L ₂) + Igα/Igβ (CD79a/b)	αβ heterodimer (or γδ) + CD3 complex
Binding sites	Two	One
Recognises	Native, free antigen — protein, sugar, lipid, nucleic acid	Only a peptide presented in an MHC groove (MHC restriction)
Epitope type	Linear and conformational	Linear peptide only
Secreted form	Yes — becomes antibody	Never secreted
Co-receptor	CD19/CD21	CD4 (MHC-II) or CD8 (MHC-I)
Somatic hypermutation	Yes	No
Gene rearrangement	V(D)J by RAG	V(D)J by RAG

γδ T cells Recognise antigen with little or no MHC restriction, sit in epithelia, and are **unusually abundant in young ruminants and pigs** — a favourite short-note in veterinary papers. **Signal 1** = TCR–peptide–MHC; **signal 2** = co-stimulation **CD28–B7**; signal 1 without signal 2 gives **anergy**. **CTLA-4** delivers the inhibitory signal.

Complement — three pathways, one lytic end

MECHANISM

CLASSICAL

Ag–Ab complex (IgM > IgG)
→ C1qrs → C4, C2

LECTIN

MBL binds mannose
→ MASP → C4, C2

ALTERNATIVE

Microbial surface, LPS —
no antibody needed; B, D, P

C3b = the great
OPSONIN (CR1)

C3a, C5a, C4a
= ANAPHYLATOXINS
mast-cell degranulation
C5a also = chemotaxin

C3 CONVERTASE → C3a + C3b

C5 CONVERTASE → C5a + C5b

MAC C5b6789 → osmotic lysis

Regulators
C1-INH, factor H & I,
DAF (CD55), CD59,
protectin — protect
host cells from lysis

Cytokines

TABLE

Cytokine	Chief source	Chief action
IL-1	Macrophage	Endogenous pyrogen — fever, acute-phase response, T-cell co-stimulation
IL-2	Th1	T-cell growth factor — clonal proliferation, NK activation
IL-4	Th2, mast cell	B-cell growth; class switch to IgE ; drives Th2
IL-5	Th2	Eosinophil growth and activation; IgA switch
IL-6	Macrophage, Th2	Acute-phase proteins, plasma-cell differentiation
IL-10, TGF-β	Treg, Th2	Immunosuppressive — dampen Th1 and macrophages
IL-12	Dendritic cell, macrophage	Drives Th1 and IFN-γ; the key Th1-polarising signal

Five functions – the standard answer

1 Cytolysis by the MAC · 2 Opsonisation (C3b) · 3 Chemotaxis (C5a) · 4 Anaphylatoxin and inflammation (C3a, C5a) · 5 Immune-complex clearance and B-cell activation (C3d–CR2). Complement is **heat-labile – 56 °C for 30 min destroys it**, which is why serum must be **inactivated** before a complement fixation test, and why guinea-pig serum is the standard source of fresh complement.

Cytokine	Chief source	Chief action
IL-17	Th17	Neutrophil recruitment, mucosal defence
IFN- α/β	Any virus-infected cell, plasmacytoid DC	Type I – antiviral ; induces the antiviral state, MHC-I up-regulation
IFN- γ	Th1, NK, Tc	Type II – the macrophage-activating factor ; up-regulates MHC-I and II; the central cytokine of cell-mediated immunity and the basis of the IFN-γ (Bovigam) test for bovine tuberculosis
TNF- α	Macrophage	Inflammation, cachexia, tumour necrosis; the mediator of endotoxic shock
Chemokines	Many	Directed leucocyte migration (IL-8/CXCL8 for neutrophils)

Properties of cytokines – write these five: **pleiotropy** (one cytokine, many effects), **redundancy** (several cytokines, one effect), **synergy**, **antagonism**, and **cascade induction**. They act at picomolar concentration, mostly **autocrine or paracrine** and only rarely endocrine, and always through a specific surface receptor.

Sheet 03 · 60-second self-test

RECALL

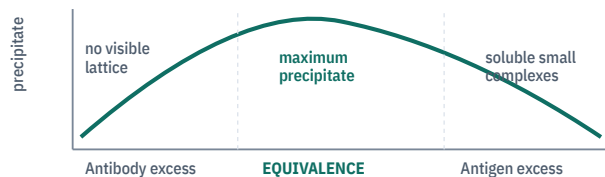
MHC-I is found on	All nucleated cells	MHC-II is found on	Professional APCs only
Light chain of MHC-I	β_2 -microglobulin	Transporter feeding MHC-I	TAP
Peptide placeholder on MHC-II	Invariant chain / CLIP	MHC of cattle / pig / dog	BoLA / SLA / DLA
MHC class III encodes	C4, C2, factor B, TNF	Co-stimulatory pair	CD28 – B7 (signal 2)
Signal 1 without signal 2 gives	Anergy	Most efficient complement fixer	IgM
Pathway needing no antibody	Alternative	Main opsonin of complement	C3b
Chemotactic complement fragment	C5a	Membrane attack complex	C5b6789
Complement inactivated by	56 °C for 30 min	Macrophage-activating cytokine	IFN- γ
Endogenous pyrogen	IL-1	T-cell growth factor	IL-2
Cytokine driving Th1	IL-12	Cytokine switching B cells to IgE	IL-4

04 Serological Reactions & Monoclonal Antibodies

t09–t10

The antigen-antibody reaction – principles that apply to every test

MECHANISM



FIVE RULES THAT ANSWER EVERY SEROLOGY QUESTION

- 1 Specificity – the reaction is lock-and-key; cross-reaction occurs when two antigens share an epitope (hetero)
 - 2 Non-covalent – held by H bonds, ionic, hydrophobic and van der Waals forces; therefore reversible
 - 3 Two stages – a rapid invisible primary binding, then a slow visible secondary stage (lattice, precipitate, clump)
 - 4 Optimal proportion – visible reaction only near equivalence; excess of either partner gives the **PROZONE** (Ab)
 - 5 Affinity vs avidity – affinity is the strength of one paratope-epitope bond; avidity is the total strength of a mu
- Titre = the reciprocal of the highest serum dilution still giving a visible reaction. A four-fold rise in paired sera 2

The tests, grouped by what you see

TABLE

Group	Test	Principle & veterinary use
Agglutination particulate antigen	Slide / rapid plate	Quick qualitative screen – pullorum disease (<i>S. Pullorum</i> whole-blood plate test), RBPT for brucellosis, blood grouping
	Tube (STAT)	Serial doubling dilutions give a titre – brucellosis, salmonellosis, leptospirosis
	Indirect / passive & latex	Soluble antigen coated on latex beads or RBCs; Coombs' test (antiglobulin) – direct for autoimmune haemolytic anaemia and neonatal isoerythrolysis , indirect for detecting free antibody

Hybridoma & monoclonal antibodies

MECHANISM

- 1 Immunise a mouse → harvest spleen B cells (short-lived, spe
- 2 Myeloma cells – immortal, but HGPRT-negative
- 3 FUSE with PEG (or electrofusion) → hybridoma
- 4 Select in HAT medium – aminopterin blocks the de novo pathway; only cells with the B-cell HGPRT survive the
- 5 Screen by ELISA → clone by limiting dilution
- 6 Expand in culture or as ascites → monoclonal antibody

Feature	Polyclonal	Monoclonal
Source	Immunised animal's serum	A single hybridoma clone
Specificity	Many epitopes	One epitope

Group	Test	Principle & veterinary use
Precipitation soluble antigen	Ring / tube	Ascoli thermoprecipitation test for anthrax in hides and decomposed tissue
	AGID / AGPT (Ouchterlony)	Double diffusion in gel; read the lines of identity, partial identity or non-identity . Used for EIA (Coggins test) , bluetongue, IBR, avian leucosis, John's
	SRID (Mancini) & immunoelectrophoresis	SRID: antibody in the gel, antigen in a well; the square of the ring diameter is proportional to antigen concentration — used to quantify immunoglobulins and assess failure of passive transfer in calves and foals . Rocket electrophoresis and counter-immunoelectrophoresis are the accelerated versions
Complement fixation	CFT	Two systems: test system (Ag + serum + complement), then the indicator system (sheep RBC + haemolysin) . No haemolysis = POSITIVE (complement was used up); haemolysis = negative. Serum must be inactivated at 56 °C/30 min. Used for glanders, brucellosis, CBPP and John's ; the WOAH standard for international trade
Neutralisation	SNT / VNT · toxin neutralisation	Antibody blocks infectivity in cell culture or in a mouse; the gold standard for measuring protective antibody and for rabies (FAVN/RFFIT)
Haemagglutination	HA and HI	Some viruses agglutinate RBCs directly (NDV, influenza, IBDV). HA titrates the virus; HI titrates the antibody — the routine flock-monitoring test for Newcastle disease vaccination
Labelled assays	ELISA	Enzyme (HRP or ALP) + substrate → colour, read at a set wavelength. Indirect (detects antibody), sandwich (detects antigen), competitive/blocking (used in DIVA testing), capture . High throughput, objective, the workhorse of national disease-surveillance programmes
	FAT / IFAT · RIA · immunochromatography	Direct FAT on brain impressions is the gold standard for rabies diagnosis . RIA uses ¹²⁵ I — most sensitive but radioactive. Lateral-flow strips are the pen-side format

Feature	Polyclonal	Monoclonal
Batch consistency	Varies	Identical for ever
Cross-reaction	More	Minimal
Cost & time	Cheap, quick	Expensive, slow
Best for	Precipitation, general detection	Diagnostics, typing, therapy, purification

Applications Serotyping FMD and bluetongue viruses, **DIVA** and blocking ELISAs, immunoaffinity purification, pregnancy and hormone kits, tumour imaging, immunotherapy, and **immunotoxins** ("magic bullets"). Chimeric, humanised and fully human mAbs (suffix **-mab**) reduce the anti-mouse antibody response.

Sheet 04 · 60-second self-test

RECALL

False negative from antibody excess	Prozone phenomenon	Definition of titre	Reciprocal of the highest reactive dilution
Proof of recent infection	Four-fold rise in paired sera	Antigen state in agglutination	Particulate
Antigen state in precipitation	Soluble	Ouchterlony technique	Double diffusion (AGID)
Coggins test detects	Equine infectious anaemia (AGID)	Test to quantify immunoglobulin	SRID (Mancini)
Positive CFT is read as	NO haemolysis	Indicator system of CFT	Sheep RBC + haemolysin
Antibody test for NDV in flocks	HI test	Gold standard for rabies diagnosis	Direct FAT on brain smear
Anthrax test on decomposed tissue	Ascoli thermoprecipitation	Antiglobulin test	Coombs'
Fusing agent in hybridoma	Polyethylene glycol	Selection medium for hybridomas	HAT — hypoxanthine, aminopterin, thymidine
Enzyme deficiency in myeloma cells	HGPRT	Monoclonal antibody discovered by	Köhler & Milstein, 1975

The four types of hypersensitivity – Coombs & Gell

TABLE

Type	Name	Mediator	Onset	Mechanism	Veterinary examples
I	Anaphylactic / immediate	IgE on mast cells & basophils	Seconds to 30 min	Re-exposure cross-links IgE on the mast cell → degranulation → histamine, leukotrienes, prostaglandins, PAF → smooth-muscle contraction, vasodilatation, oedema	Systemic anaphylaxis after vaccine, penicillin or serum; shock organ differs by species — lung in cattle, guinea pig and man; LIVER (hepatic veins) in the dog; lung + intestine in the horse and pig . Also atopic dermatitis, allergic rhinitis, milk allergy, insect-bite and sweet itch in horses. Treat with adrenaline , antihistamines, corticosteroids
II	Cytotoxic	IgG or IgM against a cell-surface antigen	Minutes to hours	Antibody binds the cell → complement lysis , opsonisation and phagocytosis, or ADCC by NK cells	Neonatal isoerythrolysis in foals, mules, piglets and kittens ; incompatible blood transfusion; autoimmune haemolytic anaemia and thrombocytopenia (Coombs' positive); myasthenia gravis; drug-induced (penicillin acting as a hapten on RBCs)
III	Immune complex (Arthus)	Soluble Ag–Ab complexes + complement	4–8 hours	Complexes formed in slight antigen excess deposit in vessel walls, glomeruli, joints and skin → complement activation → C5a → neutrophil influx → enzyme release → vasculitis and necrosis	Arthus reaction at a repeat injection site; serum sickness ; glomerulonephritis of chronic infection (equine infectious anaemia, ADV of mink, canine pyometra); purpura haemorrhagica after strangles ; blue eye of infectious canine hepatitis; hypersensitivity pneumonitis / farmer's lung from mouldy hay
IV	Delayed type (DTH), cell-mediated	Th1 cells + macrophages — NO antibody	24–72 hours	Sensitised Th1 cells re-meet antigen → IFN- γ → macrophage accumulation and activation → induration, and with persistent antigen a granuloma	Tuberculin test (the classic — read at 72 h), johnin and mallein tests, contact dermatitis (nickel, rubber, chemicals), graft rejection, granulomatous response to intracellular pathogens. Transferable by lymphocytes but NOT by serum — this experiment is the definition

Memory hook Anaphylactic · Cytotoxic · Immune-complex · Delayed = "**ACID**". Types I, II and III are **antibody-mediated and transferable by serum**; type IV is **cell-mediated and transferable only by cells**. A fifth, **stimulatory** type (antibody against a receptor, as in Graves' disease) is sometimes added.

Tolerance & autoimmunity

RAPID-FIRE

Immunological tolerance	Specific unresponsiveness to an antigen that would otherwise provoke a response. Induced most easily in the foetus or neonate , by very high or very low antigen doses, by soluble non-aggregated antigen and by the oral or intravenous route
Central tolerance	In thymus and bone marrow. Negative selection — clonal deletion of strongly self-reactive lymphocytes; AIRE drives expression of peripheral self-antigens in the thymus. Also receptor editing in B cells
Peripheral tolerance	Anergy (signal 1 without signal 2), clonal deletion , suppression by Treg (IL-10, TGF- β , CTLA-4), and immunological ignorance of sequestered antigens

Immunity to microbes — the table that answers "how does the body handle X?"

TABLE

Agent type	Chief protective mechanism	Evasion strategy to mention
Extracellular bacteria	Antibody — opsonisation (IgG + C3b), complement lysis, toxin neutralisation , blocking adhesion (IgA); neutrophils	Capsule, protein A, IgA protease, antigenic variation, biofilm

Classic demonstration	Freemartin cattle and dizygotic twin calves sharing a placental circulation become permanent blood chimeras and tolerant of each other's tissue — the natural experiment. Medawar's neonatal skin-graft work made the same point
How autoimmunity arises	1 Release of sequestered antigen (lens, sperm, myelin, thyroglobulin) after trauma or infection · 2 Molecular mimicry — a microbial epitope resembles self (<i>Campylobacter</i> and Guillain-Barré) · 3 Alteration of self by drugs or haptens · 4 Polyclonal activation by superantigens or LPS · 5 Failure of Treg or of AIRE · 6 Aberrant MHC-II expression on non-APCs · 7 Genetic (MHC haplotype) and hormonal (female bias) predisposition
Veterinary autoimmune diseases	Autoimmune haemolytic anaemia and immune-mediated thrombocytopenia (dog) · Pemphigus foliaceus (dog, horse — commonest autoimmune skin disease) · Systemic lupus erythematosus (dog, ANA positive) · Equine recurrent uveitis (moon blindness, leptospiral mimicry) · Myasthenia gravis (anti-AChR) · Lymphocytic thyroiditis (dog) · Rheumatoid arthritis · Autoimmune orchitis and thyroiditis
Diagnosis	Coombs' test, ANA, LE-cell test, direct immunofluorescence on a skin biopsy, response to immunosuppression

Agent type	Chief protective mechanism	Evasion strategy to mention
Intracellular bacteria TB, brucellosis, listeriosis	Cell-mediated immunity — Th1, IFN-γ-activated macrophages, CD8 Tc, granuloma formation. Antibody is of little value	Blocking phagosome-lysosome fusion, escape to the cytosol, resisting the respiratory burst, inhibiting MHC expression
Viruses	Interferon α/β and NK cells early; then neutralising antibody (blocks entry, prevents spread and reinfection) and CD8 Tc (clears infected cells); IgA at mucosal surfaces	Antigenic drift and shift, latency, downregulation of MHC-I, virokinases and viroceptors, infection of lymphocytes (immunosuppression by BVDV, CDV, IBDV, FIV)
Fungi	Chiefly cell-mediated — Th1 and Th17, activated macrophages and neutrophils	Capsule (<i>Cryptococcus</i>), melanin, thermotolerance, morphological switching
Parasites (helminths)	Th2 — IgE, eosinophils, mast cells , ADCC by eosinophils, mucus and "weep and sweep"	Antigenic variation, immune-complex decoys, glycan shielding, immunomodulation of the host

The one-sentence rule that examiners want: antibody handles what is outside the cell; T cells handle what is inside it. Everything above is a corollary of that sentence.

Vaccines

TABLE

Type	Advantages / drawbacks	Veterinary examples
Live attenuated	+ Strong, long, single dose; induces both humoral and cell-mediated immunity and local IgA; cheap – Risk of reversion to virulence , residual virulence in pregnant or immunosuppressed animals, needs a cold chain , may be neutralised by maternal antibody	Brucella S19 and RB51, anthrax Sterne 34F₂ , ND F1/LaSota, PPR Sungri/96, RD vaccines, IBR, canine distemper, sheep pox, BCG
Killed / inactivated (bacterin)	+ Safe, stable, no reversion – Weaker, chiefly humoral , needs an adjuvant and boosters	HS alum-precipitated and oil-adjuvant bacterins , FMD trivalent, rabies (cell-culture), leptospiral bacterin, killed ND
Toxoid	Exotoxin treated with 0.3–0.4% formalin at 37 °C — antigenic, not toxic	Tetanus toxoid , clostridial multi-component (enterotoxaemia), botulinum
Subunit / recombinant	Only the protective antigen; very safe, DIVA-friendly; expensive, needs adjuvant	FMD VP1, <i>E. coli</i> K99 pilus vaccine, feline leukaemia gp70, hepatitis B in man

Sheet 05 · 60-second self-test

RECALL

Prompt	Answer
Classified hypersensitivity into four types	Coombs & Gell
Immunoglobulin of type I	IgE
Shock organ in the dog	Liver (hepatic veins)
Drug of choice in anaphylaxis	Adrenaline
Type of neonatal isoerythrolysis	Type II, cytotoxic
Arthus reaction is type	Type III
Tuberculin reaction is type	Type IV — read at 72 h
Type transferable only by cells	Type IV
Natural example of tolerance	Freemartin / twin-calf chimerism
Thymic gene enforcing self-tolerance	AIRE
Commonest autoimmune skin disease of dogs	Pemphigus foliaceus
Immunity to intracellular bacteria	Cell-mediated (Th1, IFN-γ)
Formalin concentration for toxoid	0.3–0.4% at 37 °C
DIVA vaccine example	gE-deleted IBR · <i>Brucella</i> RB51
Commonest cause of vaccine failure in young	Maternal antibody interference

19 / 19 syllabus topics Paper II · Unit 4 Full pass ≈ 25 min

Night-before: the six diagrams + all self-tests

Type	Advantages / drawbacks	Veterinary examples
Vectored	A harmless virus carries the protective gene; good CMI	Vaccinia- and canarypox-vectored rabies for wildlife baiting, HVT-vectored ND and IBD in poultry
DNA / mRNA	Plasmid or mRNA encoding the antigen; stable, rapid to design, induces CMI	West Nile in horses (licensed DNA vaccine); mRNA platforms
Marker / DIVA	Allows Differentiating Infected from Vaccinated Animals — essential in an eradication programme	gE-deleted IBR, Brucella RB51 , FMD NSP-based companion ELISA
Autogenous	Made from the organism isolated on that very farm	Herd-specific mastitis and colibacillosis vaccines

Failure of vaccination – list six 1

Break in the **cold chain** or expired vaccine · **2 Interference by maternal antibody** in the young (why puppies need a series to 16 weeks) · **3** Wrong route, dose or timing · **4** Immunosuppression — stress, malnutrition, parasitism, concurrent BVD/IBD/CDV, corticosteroids · **5 Antigenic mismatch** between the vaccine and the field strain (FMD serotypes, ND genotypes) · **6** Genetic non-responders; and use of disinfectant on the needle or in the drinking water for live vaccines.